

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

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Abstract—Modern supply chain management relies heavily on data-driven decision-making to optimize logistics, minimize delays, and maximize profitability. This paper presents a comprehensive data analytics framework and interactive dashboard developed during an industrial internship at Unified Mentor Private Limited to evaluate the factory-to-customer shipping route efficiency for the Nassau Candy Distributor. Utilizing a historical dataset of over 10,000 shipment records, the study systematically identifies logistical bottlenecks, evaluates factory dispatch performance, and assesses geographic transit delays. The computational methodology leverages Python, specifically employing Pandas for rigorous data preprocessing and feature engineering, alongside Streamlit and Plotly for the deployment of a highly interactive visual interface. Key findings reveal a systemic Average Route Efficiency Score of 43.5/100 and a baseline delay rate of 33.1%. Furthermore, empirical analysis identifies “The Other Factory” and “Standard Class” shipping as optimal logistical nodes, while exposing severe inefficiencies originating from the “Sugar Shack” facility and “First Class” shipping modes. By formalizing a customized Route Efficiency Scoring framework and mapping geographic friction points, this research provides actionable executive recommendations to restructure distribution networks, ultimately contributing to enhanced operational scalability and customer satisfaction.

Index Terms—Supply Chain Analytics, Route Efficiency, Data Visualization, Streamlit, Logistics Optimization, Feature Engineering, Interactive Dashboards.

I. INTRODUCTION

The global logistics sector is characterized by complex, multidimensional networks where factory-to-customer routing directly dictates operational profitability and consumer retention. In distribution environments like that of the Nassau Candy Distributor, minor inefficiencies in dispatch protocols, transit modes, or regional routing compound rapidly, leading to cascaded supply chain failures. Traditional, static reporting mechanisms are increasingly insufficient for managing these dynamic networks, as they fail to provide the granular, data-driven visualization required by supply chain executives to enact systemic corrections.

This research, conducted as an enterprise data analytics project in collaboration with Unified Mentor Private Limited, addresses the critical need for interactive logistics analytics by developing a comprehensive data pipeline and dashboard. The primary objective is to dissect the operational performance of individual factories, shipping modes, and regional transit corridors through historical shipment analysis. By transforming raw, unstructured shipment logs into a centralized analytical

engine, this project facilitates the identification of geographic bottlenecks and formulates a quantitative baseline for route efficiency.

II. LITERATURE REVIEW

The integration of Business Intelligence (BI) into supply chain management has been extensively documented. Early logistics frameworks relied heavily on enterprise resource planning (ERP) systems that prioritized inventory tracking over transit optimization [1]. Recent advancements, however, have shifted focus toward interactive visualization and descriptive analytics. Studies emphasize that visual analytics tools significantly reduce cognitive load for decision-makers, allowing for the rapid identification of outliers in complex logistical datasets [2].

In the domain of logistics optimization, data-driven methodologies are crucial for minimizing lead times and improving Service Level Agreement (SLA) compliance [3]. Furthermore, the utilization of Python in supply chain environments has revolutionized data processing [4]. While traditional enterprise BI solutions (e.g., Tableau, PowerBI) offer robust architectures, they often lack the rapid prototyping agility provided by open-source, component-based frameworks like Streamlit [5]. This paper bridges the existing gap in the literature by demonstrating the efficacy of a lightweight, Python-centric web application in generating enterprise-grade supply chain insights and quantifying transit efficiencies through a customized mathematical scoring framework.

III. PROBLEM STATEMENT

Despite possessing extensive historical shipment data, the Nassau Candy Distributor faces persistent challenges in optimizing its factory-to-customer delivery pipeline. The organization struggles with an unquantified delay percentage and lacks a standardized metric to evaluate the efficacy of specific shipping routes. Consequently, executive stakeholders are unable to accurately attribute logistical failures to distinct factories, shipping modes, or state-level geographical bottlenecks. This operational opacity necessitates the development of an analytical tool capable of parsing multidimensional historical data to isolate underperforming sectors and recommend data-backed restructuring strategies.

IV. OBJECTIVES

The core objectives of this research are multi-faceted, aiming to provide a holistic operational overview:

- **Factory Performance Analysis:** Evaluate and rank dispatch facilities based on processing speed, order volume, and reliability.
- **Shipping Delay Quantification:** Calculate global and regional delay percentages to establish definitive operational baselines.
- **Route Efficiency Calculation:** Formulate and apply a standardized scoring framework to quantify the performance of all active transit paths.
- **Geographic Bottleneck Identification:** Map state-level transit times to isolate infrastructural and regional friction points.
- **Shipping Mode Optimization:** Compare the empirical efficacy of various transit classifications (e.g., First Class vs. Standard).
- **Executive Recommendations:** Translate analytical findings into actionable business strategies to mitigate supply chain friction.

V. DATASET DESCRIPTION

The analysis is predicated on a proprietary dataset, `Nassau_Candy_Distributor.csv`, sourced from historical logistical records provided during the Unified Mentor internship.

TABLE I
DATASET SCHEMA AND MAJOR ATTRIBUTES

Feature Name	Data Type	Description
Order Date	Datetime	Timestamp of initial customer order.
Ship Date	Datetime	Timestamp of physical factory dispatch.
Factory	Categorical	Originating facility for merchandise.
Ship Mode	Categorical	Logistics classification (e.g., Standard).
State	Categorical	Specific state-level destination.
Customer	String	Unique identifier for receiving entity.
Sales	Float	Monetary value of transaction.
Profit	Float	Net profit after operational costs.
Quantity	Integer	Total volume of product dispatched.

The dataset encompasses over 10,000 unique shipment records and features 12 primary attributes (including engineered routing keys), spanning multiple historical operational cycles. This volume provides a statistically significant sample size for robust variance analysis and trend identification.

VI. METHODOLOGY

The computational architecture relies exclusively on a modern Python data stack, ensuring high scalability and algorithmic efficiency.

A. Data Cleaning and Preprocessing

Raw logistics data is inherently noisy and requires rigorous standardization. Pandas and NumPy were utilized for vectorized data manipulations.

- **Missing Value Handling:** Null values within critical geographic fields were systematically addressed. Rows lacking primary destination keys were dropped to maintain spatial mapping integrity, while numerical missing values in financial columns were imputed using regional median baselines.
- **Date Conversion:** `Order Date` and `Ship Date` were ingested as string literals and parsed into ISO 8601 standardized Datetime objects, a strict prerequisite for temporal arithmetic.

B. Feature Engineering and Route Mapping

To generate actionable supply chain KPIs, new analytical vectors were engineered from the baseline schema.

- **Route Creation:** A composite string, `Shipping Route`, was generated by concatenating the `Factory` and `State` fields (e.g., "Sugar Shack → New Jersey"), allowing for granular, path-specific analysis.
- **Lead Time Calculation:** Transit speed was calculated as the absolute temporal difference between dispatch and order instantiation:

$$L_i = T_{ship,i} - T_{order,i} \quad (1)$$

where L_i represents the Lead Time for order i in days.

- **Delay Status:** A boolean matrix was generated by evaluating L_i against standard Service Level Agreements (SLA_{target}):

$$D_i = \begin{cases} 1, & \text{if } L_i > SLA_{target} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

C. Route Efficiency Score Formula

To establish a quantifiable leaderboard of logistics paths, a customized Route Efficiency Score (E_{route}) was formulated. The score normalizes performance on a 0 to 100 scale, inversely penalizing routes for extended average lead times (L_{avg}) and high historical delay probabilities (P_{delay}).

Let L_{max} denote the maximum tolerable network lead time. The framework is defined as:

$$E_{route} = \max \left(0, 100 \times \left(1 - \left(0.6 \frac{L_{avg}}{L_{max}} + 0.4 P_{delay} \right) \right) \right) \quad (3)$$

This weighted equation prioritizes structural transit speed (60% weight) while heavily penalizing paths with a high propensity for erratic SLA failures (40% weight).

D. Dashboard Development and Visualization Workflow

The deployment phase culminated in an interactive web application built via Streamlit. Plotly's WebGL-based rendering engine was employed to generate interactive geographic choropleth maps and statistical charts. The dashboard architecture segregates spatial data from categorical performance metrics,

enabling stakeholders to apply dynamic filters (e.g., Date Range, Factory, Ship Mode) that instantaneously recalculate all underlying DataFrames without page reloads.

VII. EXPERIMENTAL RESULTS

The execution of the analytical pipeline over the 10,000+ records yielded highly distinct operational metrics, summarized in Table II.

TABLE II
GLOBAL LOGISTICS KEY PERFORMANCE INDICATORS (KPIs)

Metric	Recorded Value
Total Orders	10,194
Total Sales	\$141,784
Total Profit	\$93,443
Average Route Efficiency Score	43.5 / 100
Global Delay Percentage	33.1%

A. Lead Time Data Characteristics

It is imperative to note that the dashboard reports an exceptionally high average lead time of approximately 1,320 days. This value is derived directly from the temporal attributes of the provided dataset, where recorded Ship Dates extend significantly beyond the initial Order Dates. Rather than artificially modifying or synthesizing new source values, the analytical framework faithfully reflects the raw dataset to maintain absolute data integrity. This anomaly highlights the presence of synthetic or historically staggered timestamps within the organizational database.

B. Performance Extremes and Bottlenecks

Analysis of specific categorical dimensions revealed extreme polarities in network performance:

- **Factory Performance:** “The Other Factory” emerged as the most optimal node, demonstrating the highest throughput and lowest correlation with shipping delays. Conversely, the “Sugar Shack” facility acted as a statistically significant bottleneck.
- **Shipping Modes:** Paradoxically, “Standard Class” exhibited the highest reliability and efficiency score. “First Class” shipping demonstrated the highest variance in lead times and consistent SLA failures.

As shown in geographic tracking, regional routing extremes were mathematically isolated. West Virginia recorded the highest state-level delays, while Maine operated with the highest regional efficiency. Furthermore, the fastest established route (Secret Factory → Nebraska) drastically outperformed the network’s slowest path (Sugar Shack → New Jersey).

VIII. BUSINESS INSIGHTS & RECOMMENDATIONS

The interactive dashboard serves as a foundational tool for data-backed business restructuring. The empirical results expose a systemic network failure, evidenced by the 33.1% global delay rate. Based on the findings, the following professional recommendations are proposed:

- **Reassess “First Class” Logistics Contracts:** The analytical failure of “First Class” shipping indicates a misalignment in carrier resource allocation. It is recommended to suspend or renegotiate SLA terms with carriers handling premium shipments and temporarily divert critical volume to the more reliable “Standard Class” network.
- **Audit “Sugar Shack” Operations:** The extreme delays routing to New Jersey dictate an immediate need for systemic management restructuring at the Sugar Shack facility. Alternatively, East Coast fulfillment should be rerouted to alternative nodes.
- **Standardize Workflow Procedures:** Conduct an internal workflow analysis at “The Other Factory” to identify standard operating procedures that can be exported to underperforming facilities.
- **Geographic Network Rerouting:** Implement alternative last-mile carrier networks for deliveries destined for the Appalachian region (specifically West Virginia) to bypass current infrastructural friction points.

IX. LIMITATIONS

While robust in its historical analysis, this research is subject to specific constraints:

- **Historical Data Constraint:** The dashboard visualizes historical batch records and lacks live, real-time API tracking.
- **Synthetic Date Values:** As noted, the dataset contains synthetic or legacy shipment dates, resulting in an inflated average lead time of 1,320 days, which limits temporal realism without altering the data’s raw state.
- **Absence of Exogenous Variables:** The analysis lacks external datasets such as localized weather events, vehicular traffic data, or specific carrier identities (e.g., FedEx vs. UPS), limiting the ability to ascertain the exact exogenous causes of transit delays.
- **Missing Granularity:** The dataset lacks warehouse-level operational metrics, such as internal picking and packing times.

X. FUTURE SCOPE

To scale the operational impact of this framework, future iterations will focus on advanced technical enhancements:

- **Machine Learning Integration:** Deploy predictive algorithms (e.g., XGBoost, Random Forest) to forecast probable transit delays before dispatch occurs, based on historical route efficiencies.
- **Route Optimization Algorithms:** Implement graph-based optimization algorithms (such as Dijkstra’s or A* search) to mathematically suggest optimal rerouting protocols during carrier failures.
- **Real-Time API Tracking:** Transition the architecture from static CSV ingestion to a live PostgreSQL database fed by carrier APIs, enabling automated alerts and real-time dashboard updates.

XI. CONCLUSION

The Factory-to-Customer Shipping Route Efficiency Analysis effectively demonstrates the transformative capability of data analytics in supply chain operations. By deploying a comprehensive Python-based architecture utilizing Pandas and Streamlit, this research successfully replaced opaque logistical reporting with an interactive, quantitative analytical framework. The identification of a 33.1% baseline delay rate, the formalization of the Route Efficiency Score, and the empirical isolation of the “Sugar Shack” and “First Class” shipping bottlenecks provide the Nassau Candy Distributor with a definitive blueprint for structural optimization. Ultimately, this project validates that structured feature engineering and interactive data visualization are indispensable for enhancing logistical scalability and operational reliability.

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